

Vectorization — Speeding Up Computations in Deep Learning

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What is Vectorization?

- Replace explicit element-wise loops with **array/matrix operations**.
- Lets NumPy/BLAS use CPU SIMD and GPU kernels for **massive speedups**.
- Crucial for training on large datasets and deep models.

Example: Dot Product $z = W^T X + b$

Non-vectorized

```
import numpy as np
z = 0.0
for i in range(n_x):
    z += W[i] * X[i]
z += b
```

Vectorized

```
import numpy as np
z = np.dot(W, X) + b
```

- Same result; vectorized version avoids Python loop overhead.

Logistic Regression (Single Example)

Non-vectorized

```
# W: (n_x,), X: (n_x,)
z = 0.0
for i in range(n_x):
    z += W[i] * X[i]
z += b
A = 1/(1 + np.exp(-z))
```

Vectorized

```
z = np.dot(W, X) + b
A = 1/(1 + np.exp(-z)) # sigma(z)
```

Mini-batch Vectorization (m Examples)

- Stack examples as columns: $X \in \mathbb{R}^{n_x \times m}$, labels $Y \in \mathbb{R}^{1 \times m}$, parameters $W \in \mathbb{R}^{n_x}$.

```
# X: (n_x, m), W: (n_x,), b: scalar
Z = np.dot(W.T, X) + b          # (1, m)
A = 1/(1 + np.exp(-Z))         # (1, m)
# Gradients for logistic loss
dZ = A - Y                     # (1, m)
dW = (1/m) * np.dot(X, dZ.T)   # (n_x, 1)
db = (1/m) * np.sum(dZ)        # scalar
W = W - alpha * dW.ravel()
b = b - alpha * db
```

Why Vectorization is Fast

- NumPy calls optimized BLAS/LAPACK: **SIMD** (Single Instruction, Multiple Data).
- Fewer Python-level ops; heavy math in compiled C/Fortran kernels.
- Enables GPU acceleration with compatible libraries.

Rule of Thumb

- Avoid explicit for-loops over samples/features.
- Prefer `np.dot`, `np.matmul`, broadcasting, and reductions (`sum`, `mean`).
- Write equations first; implement them as whole-array ops.

Summary

Concept	Benefit
Vectorization	Eliminates Python loops
BLAS/NumPy	Uses parallel hardware (SIMD/GPU)
Mini-batch math	Clean, short, fast code